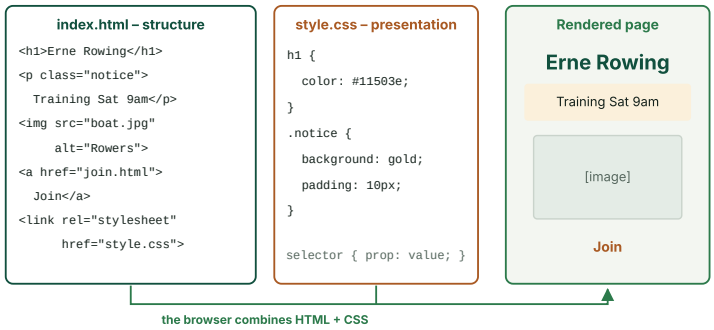
HTML and CSS

FD18
FACT FILE

Specification: students should be able to

- Explain how web pages are created using Hypertext Markup Language (HTML), including the html, head and body structure
- Describe the purpose of common HTML tags: paragraph, headings, line break, image, anchor (hyperlink), ordered list and unordered list
- Describe the purpose of tag attributes, empty and container tags, character entities and simple tables and forms
- Explain how Cascading Style Sheets (CSS) are used to control the presentation of web pages, using selectors, properties and values
- Compare inline, internal (global) and external style sheets and explain how inheritance and precedence resolve style conflicts
- Explain the benefits of separating content (HTML) from presentation (CSS)

Key words: HTML, tag, container tag, empty tag, attribute, hyperlink, anchor, alt text, character entity, CSS, selector, declaration, class, id, inline style, external style sheet, inheritance, precedence



One external style sheet styles every page: consistent, quick to change, cached by the browser

HTML provides structure and content, an external CSS style sheet provides presentation, and the browser combines them into the rendered page

How a web page is built with HTML

A website is a set of linked web pages, and each page is a text file written in Hypertext Markup Language. HTML does not decide how things look; it marks up the content so that a browser knows what each part is. Tags are keywords written inside angle brackets, and an element is usually an opening tag, some content and a closing tag in which a forward slash comes before the keyword. A modern page begins with the declaration <!DOCTYPE html>, followed by an html element containing two sections. The head holds information about the document as a whole: the title shown on the browser tab, the character set, and links to style sheets or scripts. The body holds everything the user actually sees, such as headings, paragraphs, lists, images, tables, forms and hyperlinks. When a user requests a page, the browser downloads the HTML, reads the tags from top to bottom, fetches any linked files such as images and style sheets, and renders the result in the window. Because HTML describes meaning rather than appearance, the same page can be displayed sensibly on a large monitor, a phone or a screen reader, with Cascading Style Sheets supplying the visual design.

Tags, attributes and text formatting

Most HTML elements use container tags, which come as a pair and wrap content, for example a paragraph or an anchor. A few use empty tags, which stand alone because they have no content to enclose; the line break, image, link and meta tags are common examples. Attributes refine an element and are written inside the opening tag as a name, an equals sign and a value in quotation marks, such as the href of a link or the src and alt of an image. Tag names are not case sensitive, but writing them consistently in lower case makes code easier to read. Browsers treat any run of spaces, tabs and new lines in the source as a single space, so pressing Enter in the code does not create a new line on screen; the br tag does that, while the p tag starts a new paragraph with a gap above it. Headings run from h1, the most important and largest, down to h6. Characters that are reserved in HTML, such as the less-than sign and the ampersand, or that are awkward to type, such as the copyright symbol or accented letters, are written as character entities beginning with an ampersand and ending with a semicolon.

Lists, hyperlinks, images and tables

An unordered list is enclosed in ul tags and shows a bullet before each item, while an ordered list uses ol tags and numbers its items; in both, each item sits inside li tags. The type attribute of an ordered list changes the numbering style, so a value of A gives capital letters and I gives Roman numerals. Hyperlinks are created with the anchor tag: the href attribute holds the destination and the text between the tags is what the user clicks. A relative address such as join.html points to another page on the same site, an absolute address beginning with https points anywhere on the web, and a mailto address opens a new email instead of a page. Images are added with the empty img tag. Its src attribute gives the file name, including the folder path if the image is stored elsewhere, and its alt attribute provides a text alternative that appears if the image fails to load and is read aloud by screen readers. Wrapping an img inside an anchor makes the picture clickable. Tables, which the specification does not require you to write but which are widely used, are built from table, caption, thead, tbody, tr, th and td elements.

Forms

Forms let a page collect data from users, for example a booking for a training session or a request for a brochure. The form element gives two important attributes: action names the server-side script that will receive the data, and method states how it is sent. With post, the data travels in the body of the HTTP request rather than being added to the address, which is more suitable for personal details. Inside the form, input elements have a type that controls both their appearance and what the browser accepts: text, email, number, password, date, checkbox, radio and submit are the most common. Each control should have a label, linked by matching the label's for attribute to the input's id, so that clicking the label selects the field and screen readers announce what it is for. HTML attributes provide simple validation in the browser before anything is sent: required acts as a presence check, min and max as a range check, maxlength as a length check, and pattern as a format check. These checks give quick feedback, but a user can bypass them, so the server must always validate the data again, as the next lesson explains.

CSS rules, selectors and colours

Cascading Style Sheets describe how HTML elements are presented. A CSS rule starts with a selector, which picks out the elements to be styled, followed by braces containing one or more declarations. Each declaration is a property, a colon and a value, and declarations are separated by semicolons. An element selector such as p applies to every paragraph. A class selector, written with a full stop, applies to every element given that class, so it suits styles that repeat, such as a highlighted fixture. An id selector, written with a hash, applies to the single element with that id, because an id must be unique within a page. Selectors can be grouped with commas or combined, so nav a means links that sit inside the navigation bar. Colours can be given in three ways: by one of the standard colour names, by rgb() with red, green and blue values from 0 to 255, or by a hash followed by six hexadecimal digits, two for each primary from 00 to FF. For example, green, rgb(0, 128, 0) and #008000 all give the same colour, because 128 in denary is 80 in hexadecimal. CSS uses the American spelling color, and background-color fills the area behind an element.

Inline, internal and external style sheets

Style information can be placed in three locations. An inline style is written in the style attribute of a single element; it is quick for a one-off change but has to be repeated on every element that needs it, which mixes presentation into the content and makes the site hard to update. An internal, or global, style sheet is a style block in the head of a page; its rules apply to every matching element on that page, but every other page needs its own copy. An external style sheet is a separate file with a .css extension, attached to each page with an empty link tag in the head that gives rel="stylesheet" and the file name in href. External style sheets are the most flexible approach. One file controls the house style of the whole site, so changing a colour scheme or font means editing one line rather than hundreds of pages, and every page stays consistent. The HTML files become smaller and the browser downloads and caches the style sheet once, so later pages load faster. Different style sheets can also be supplied for phones, printing or users who need high contrast, all using the same HTML content.

Resolving conflicts: inheritance and precedence

The word cascading refers to the predictable rules a browser uses when more than one style could apply to an element. Inheritance means that elements nested inside another element take on many of their parent's properties, especially text properties such as color, font-family and font-size; a span inside a blue Arial paragraph is also blue Arial unless told otherwise. Not every property is inherited: borders, padding and backgrounds belong to the element they are set on. Precedence means that when the same property is defined in more than one place, the definition closest to the element wins. An inline style overrides a rule for that element in an internal or external style sheet, and a rule set directly on a child overrides the value it would have inherited from its parent. A new property on the child simply adds to the inherited ones. For example, if paragraphs are blue Arial and a span inside one has an inline style of color: red, the span text appears in red Arial, because the font is inherited but the colour is overridden. If that inline rule were removed, the text would revert to blue. Where no rule applies at all, the browser's default style is used.

T Key terms from the mark schemes

HTML
Hypertext Markup Language: the markup language that uses tags to define the structure and content of a web page, such as headings, paragraphs, lists, images and links.
`Credited: Language used to create web pages`

Container tag
An HTML tag that is used as an opening and closing pair enclosing content, such as ... or <a> ... .
`Credited: Paired tag`

Empty tag
An HTML tag that stands alone with no closing tag because it encloses no content, such as
, <img> or <link>.
`Credited: Void element / self-closing tag`

Attribute
Extra information written inside an HTML opening tag as name="value", which refines the element, such as href, src or alt.

Anchor tag
The <a> tag, which creates a hyperlink; its href attribute gives the destination URL and the enclosed content is what the user clicks.
`Credited: Hyperlink tag`

Alt text
The alt attribute of an image, giving a text description that is displayed if the image cannot load and is read aloud by screen readers.
`Credited: Alternative text`

CSS
Cascading Style Sheets: the language used to describe the presentation of web pages, such as colours, fonts, spacing and layout, separately from the content.

Selector
The part of a CSS rule that identifies which HTML elements the declarations apply to, e.g. an element name, a .class or an #id.

External style sheet
A separate .css file linked to web pages with the <link> tag so that one set of style rules controls the appearance of the whole site.
`Credited: Linked style sheet`

Inline style
A style written in the style attribute of a single HTML element, affecting only that element.

Inheritance
The way a nested (child) element takes on style properties such as colour and font from the element that contains it (its parent).

Precedence
The rule that when the same property is defined in more than one place, the definition closest to the element (e.g. an inline style) takes priority.
`Credited: Closest definition wins`

Exam focus

Questions on this topic often ask you to write short pieces of HTML or CSS, to explain what code will display, or to compare ways of applying styles. When writing HTML, close every container tag, nest elements correctly, use the exact attribute names href, src and alt, and put attribute values in quotation marks; a mark is usually given for each correct element. Changing a bulleted list to a numbered one only needs the outer tags changed from ul to ol. In CSS, remember color without a u, a colon between property and value, semicolons between declarations, a full stop for a class and a hash for an id. When asked how text will appear, work through each property separately: say which value is inherited, which is overridden and why, using the words inheritance and precedence. Questions on the purpose of CSS should contrast it with HTML: HTML describes the content and structure, CSS describes the appearance. Questions on linked or external style sheets expect an explanation, not a list: one change in the CSS file is applied to every linked page, which keeps a house style consistent and reduces maintenance, and styles can be adapted for different devices.

? Check yourself

1. HTML elements are written using either container tags or empty tags. Describe what is meant by each type of tag and give one example of each. [4]
2. A walking club in the Mourne Mountains wants a page that shows the heading 'Upcoming walks', a sentence 'All walks start at 10am.', a numbered list of three walks (Slieve Donard, Slieve Binnian, Slieve Meelmore) and a link with the text 'Book a place' that opens booking.html. Write the HTML for the body of this page. [6]
3. Write the HTML to display the image logo.png, stored in a folder called images, so that clicking the image takes the user to index.html. Include suitable alternative text. [3]

Answers: 1. Container tag: comes as an opening and closing pair (1) which encloses content (1), e.g. ... , <a> ... , <h1> ... </h1> (1) / Empty tag: stands alone with no closing tag/no content, e.g.
, <img>, <link> (1). 2. <h1>Upcoming walks</h1> (1) / All walks start at 10am. (1) / ... used for the numbered list (1) / three <li> ... items with the correct text (1) / (1) ... Book a place with correct closing tag (1). Accept any heading level; accept ul only if numbering added by other means (not expected). 3. (1) / <img src="images/logo.png" (1) alt="[suitable description, e.g. Club logo – home page]"> (1) . The img must be inside the anchor for full marks.